IMPERIAL COLLEGE LONDON

EE4-24
EE9-CS1-3

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-TIME SYSTEMS AND COMPUTER CONTROL

Thursday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A. Astolfi

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DISCRETE-TIME SYSTEMS AND COMPUTER CONTROL

Information for candidates:

- $Z\left(\frac{1}{s}\right) = Z(1(kT)) = \frac{z}{z-1}$
- $Z\left(\frac{1}{s+a}\right) = Z\left(e^{-akT}\right) = \frac{z}{z-e^{-aT}}$
- $Z\left(\frac{1}{s^2}\right) = Z(kT) = \frac{zT}{(z-1)^2}$
- Let $F(s)$ be the Laplace transform of $f(t)$ and n a positive integer. Then

$$Z\left(e^{-nTs}F(s)\right) = Z(f(kT-nT)1(kT-nT)) = z^{-n}Z(f(kT))$$

- Transfer function of the ZOH: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Definition of the w-plane: $z = \frac{1+\frac{wT}{2}}{1-\frac{wT}{2}}$, $w = \frac{2}{T}\frac{z-1}{z+1}$
- Note that, for a given signal r (also indicated as $r(k)$, $r(kT)$ or $r(t)$), $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.
- Consider the characteristic equation $1 + K\frac{N(z)}{D(z)} = 0$. $N(z)$ has roots z_i, $i = 1,\ldots,m$, and $D(z)$ has roots p_i, $i = 1,\ldots,n$. If $n > m$ the asymptotes of the root locus intersect the real axis at $\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{n-m}$, and radiate out with angles $\theta = \pm q\frac{\pi}{n-m}$, where $q = 1,3,5,\ldots$. The break-away or break-in points are the points of the locus which are also solutions of the equation $N(z)\frac{d}{dz}D(z) - D(z)\frac{d}{dz}N(z) = 0$.

1. Consider the digital control system in Figure 1.1

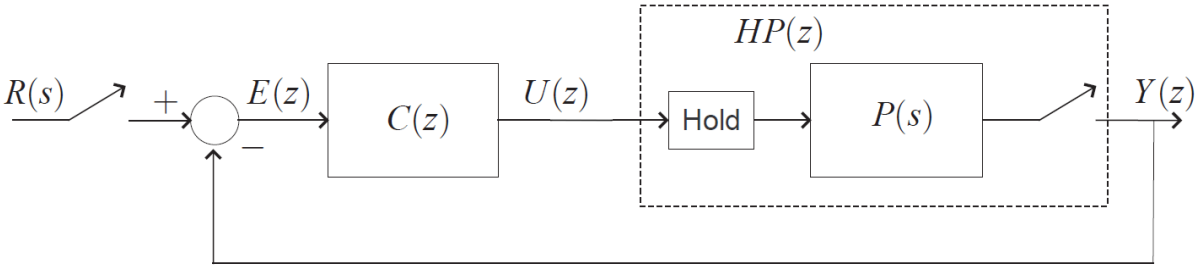


Figure 1.1 Block diagram for Question 1.

with transfer function

$$P(s) = \frac{2}{s-2}$$

and the problem of designing a feedback controller. Let $T = 0.04$ be the sampling time.

a) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to a ZOH and a sampler.

[4 marks]

b) Determine the equivalent transfer function $HP(w)$ in the w-plane.
Hint: you can use the approximation $0.03997 \approx 0.04$.

[2 marks]

c) Design, in the w-plane, a controller

$$C(w) = k\frac{w+a}{w+b}$$

such that the closed-loop system (in the w-plane) is asymptotically stable, has DC gain equal to 2 and has one pole equal to -1 and one pole equal to -2.

[8 marks]

d) Consider now the controller

$$C(w) = 0.2577\frac{w+96}{w+12.3711}$$

which has been designed in such a way that the closed-loop system (in the w-plane) has two poles in -5.

i) Compute the transfer function $C(z)$ of the discrete-time controller. Show all the steps of your work.

ii) Compute the poles of the closed-loop transfer function (in the z-domain). Show all the steps of your work.

iii) Compute the DC gain of the closed-loop transfer function (in the z-domain). Show all the steps of your work.

[6 marks]

2. This question tests three different concepts seen in the module. You must explain all the steps of your work to receive full marks.

a) Consider the solid-grey s-plane region shown in Figure 2.1. Assume that the grey region extends to negative infinity on the left-hand side. Let ω_s be the sampling frequency and let the primary strip be identified by the region between $-j\omega_s$ and $+j\omega_s$. Sketch the z-plane region that corresponds to the solid-grey s-plane region. In particular, label the 5 black dots and map them clearly on the z-plane.

[6 marks]

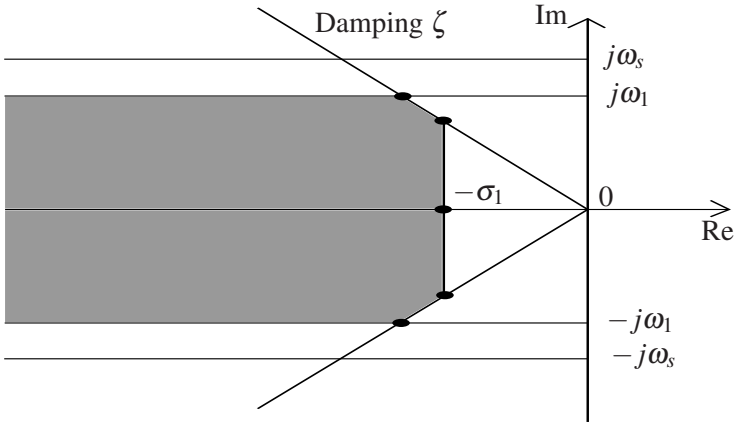


Figure 2.1 Solid-grey s-plane region for Question 2.

b) Consider the digital control system in Figure 2.2. Assume that the samplers are impulsive samplers. Determine the pulse transfer function between the input $R(z)$ and the output $Y(z)$.

[7 marks]

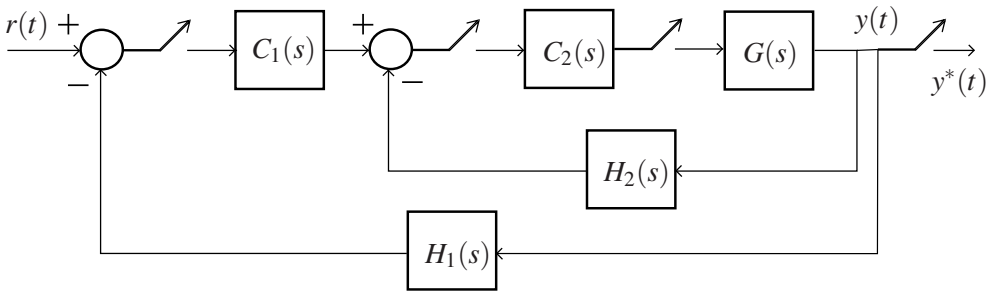


Figure 2.2 Block diagram for part c) of Question 2.

c) Consider the digital control system in Figure 2.3. Assume that the reference signal is $R(z) = 0$. Determine the pulse transfer function between the disturbance $D(z)$ and the output $Y(z)$. Explain how the gains of $C_1(z)$ and $C_2(z)$ should be selected to minimize the effect of the disturbance.

[7 marks]

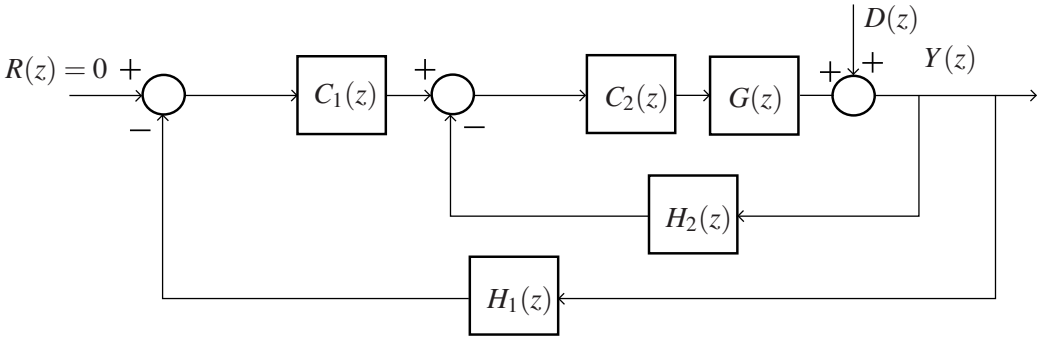


Figure 2.3 Block diagram for part d) of Question 2.

3. Consider the digital control system in Figure 3.1

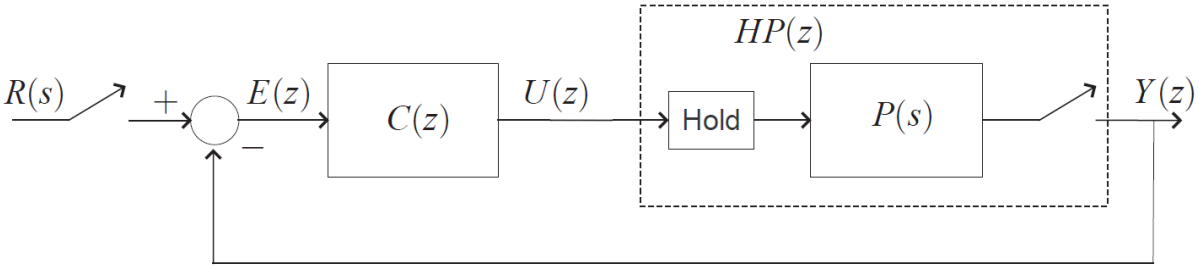


Figure 3.1 Block diagram for Question 3.

with transfer function

$$P(s) = e^{-2Ts}\frac{1}{s+1}$$

and the problem of designing a feedback controller.

a) Compute the equivalent discrete-time model $HP(z)$ for the plant interconnected to a ZOH and a sampler. Once you have found $HP(z)$ replace $T = \ln(2)$.
Hint: check the "Information for candidates".

[4 marks]

b) Let $C(z) = k$, where k is a gain to be determined. Determine the range $(k_{\min}, k^{\max})$ of values of k such that the closed-loop system is asymptotically stable. Show all the steps of your work to receive full marks.

[6 marks]

c) Sketch the root locus of $C(z)HP(z)$ for $k \geq 0$. You need to justify all your work by means of computations and written explanations: if there are asymptotes, compute their intersection point and the departure angles; compute all break-in and break-out points and indicate on the locus the points corresponding to $k_{\min}$ and $k^{\max}$. A sketch without any justification will receive 0 marks.
NB: ignore any break-away point corresponding to non-positive values of k.

[6 marks]

d) Let $\alpha > 0$. Explain how one can determine the range of values of k such that the closed-loop system has dominant poles in absolute value less than α.

[4 marks]

4. Consider the continuous-time equation

$$\dot{x}(t) = ax(t) + bu(t), \tag{4.1}$$

where a and b are constants (scalars). Recall that the state $x(t)$ of (4.1) can be expressed as

$$x(t) = e^{a(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{a(t-\tau)}bu(\tau)d\tau, \qquad t \geq t_0. \tag{4.2}$$

Consider the performance index to be minimized

$$J = \frac{1}{2}x^2(NT) + \frac{1}{2}\int_0^{NT} [qx(t)^2 + ru(t)^2]dt, \tag{4.3}$$

where q and r are constants (scalars). The objective of this exercise is to formulate the quadratic optimal control problem for a discretized control system.

a) Assume that the input $u(t)$ is sampled and fed to a zero-order hold so that $u(t)$ is constant over the interval between two consecutive sampling instants. Prove that the behaviour of the discretized system is described by

$$x(k+1) \quad = \quad fx(k) + gu(k), \tag{4.4}$$

with

$$f = e^{aT}, \qquad g = \frac{b}{a}\left(e^{aT} - 1\right).$$

Hint: use $t = (k+1)T$ and $t_0 = kT$ in (4.2). Then use the change of variable $\lambda = (k+1)T - \tau$.

[2 marks]

b) Pick $t_0 = kT$ and show that, for $kT \leq t < (k+1)T$, (4.2) can be rewritten as

$$x(t) = \xi(t-kT)x(kT) + \eta(t-kT)u(kT). \tag{4.5}$$

Compute $\xi(t-kT)$ and $\eta(t-kT)$ explicitly (solve any integral appearing in the expressions).

[6 marks]

c) The performance index (4.3) can be rewritten as

$$J = \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1}\int_{kT}^{(k+1)T} [qx(t)^2 + ru(t)^2]dt. \tag{4.6}$$

By replacing (4.5) in (4.6) show that

$$J = \frac{1}{2}x^2(NT) + \frac{1}{2}\sum_{k=0}^{N-1} [\widetilde{q}x(kT)^2 + \widetilde{m}x(kT)u(kT) + \widetilde{r}u(kT)^2]dt \tag{4.7}$$

and compute $\widetilde{q}$, $\widetilde{m}$ and $\widetilde{r}$ explicitly (solve any integral).

[8 marks]

d) Comment on the following.

i) Comment on the difference between the indexes (4.3) and (4.7)? Explain the origin of the difference.

ii) Ways to solve the obtained optimal control problem by using the solution seen in class for the problem without the cross-term involving $x(kT)$ and $u(kT)$. (Just explain the idea, there is no need to prove or do any computation).

[4 marks]